Jensen's Inequality with Convex Envelopes

April 19, 2026

The usual perspective on Jensen's inequality is that it allows to bound expectations involving convex functions. A basic observation is that it can also be useful to bound expectations involving functions *which can themselves be bounded by convex functions.*

A First Illustration

I'll illustrate this by example. Let $p > 1$, and let $F : \mathbf{R}_+ \to [0,1]$ be given by $F(t) = \exp(-t^p)$. Given a non-negative random variable X, I would like to lower-bound $\mathbf{E}[F(X)]$ in terms of $\mathbf{E}[X]$. (I forget the exact motivation for this; I think that it involved acceptance rates for the Random Walk Metropolis algorithm under different smoothness conditions). Anyways, F is non-convex for small t, and so Jensen's inequality isn't much use to us. Or is it?

Let us compute the convex envelope of F, i.e. the largest convex function G which minorises F globally. With a little bit of work, one can show that it takes the form

$$G(t) = \begin{cases} 1 - c_\star \cdot t & t \in [0, t_\star] \\ F(t) & t \in [t_\star, \infty) \end{cases}$$

where $t_\star > 0$ satisfies $(1 + p \cdot t_\star^p) \cdot \exp(-t_\star^p) = 1$, and $c_\star = p \cdot t_\star^{p-1} \cdot \exp(-t_\star^p)$. For a given p, these quantities can be computed relatively easily.

Now, since G minorises F, it is clear that $\mathbf{E}[F(X)] \geq \mathbf{E}[G(X)]$. Moreover, since G is convex, Jensen's inequality does apply, and so we can bound $\mathbf{E}[G(X)] \geq G(\mathbf{E}[X])$. Combining the two, we see that $\mathbf{E}[F(X)] \geq G(\mathbf{E}[X])$, i.e. we have a lower bound for $\mathbf{E}[F(X)]$ in terms of $\mathbf{E}[X]$.

For extra ease of use, note that for $t \in [0, t_\star]$, one can lower bound $1 - c_\star \cdot t \geq \exp(-\lambda_\star \cdot t)$, where $\lambda_\star = t_\star^{p-1}$. Using this, one can obtain the looser (but perhaps easier-to-use) bound of

$$G(t) \geq \exp\left(-t_\star^p \cdot \max\left\{\left(\frac{t}{t_\star}\right), \left(\frac{t}{t_\star}\right)^p\right\}\right).$$

One thus ends up with the general result that for nonnegative random variables X, there holds the bound

$$\mathbf{E}\left[\exp\left(-X^p\right)\right] \geq \exp\left(-t_\star^p \cdot \max\left\{\left(\frac{\mathbf{E}[X]}{t_\star}\right), \left(\frac{\mathbf{E}[X]}{t_\star}\right)^p\right\}\right).$$

The General Case

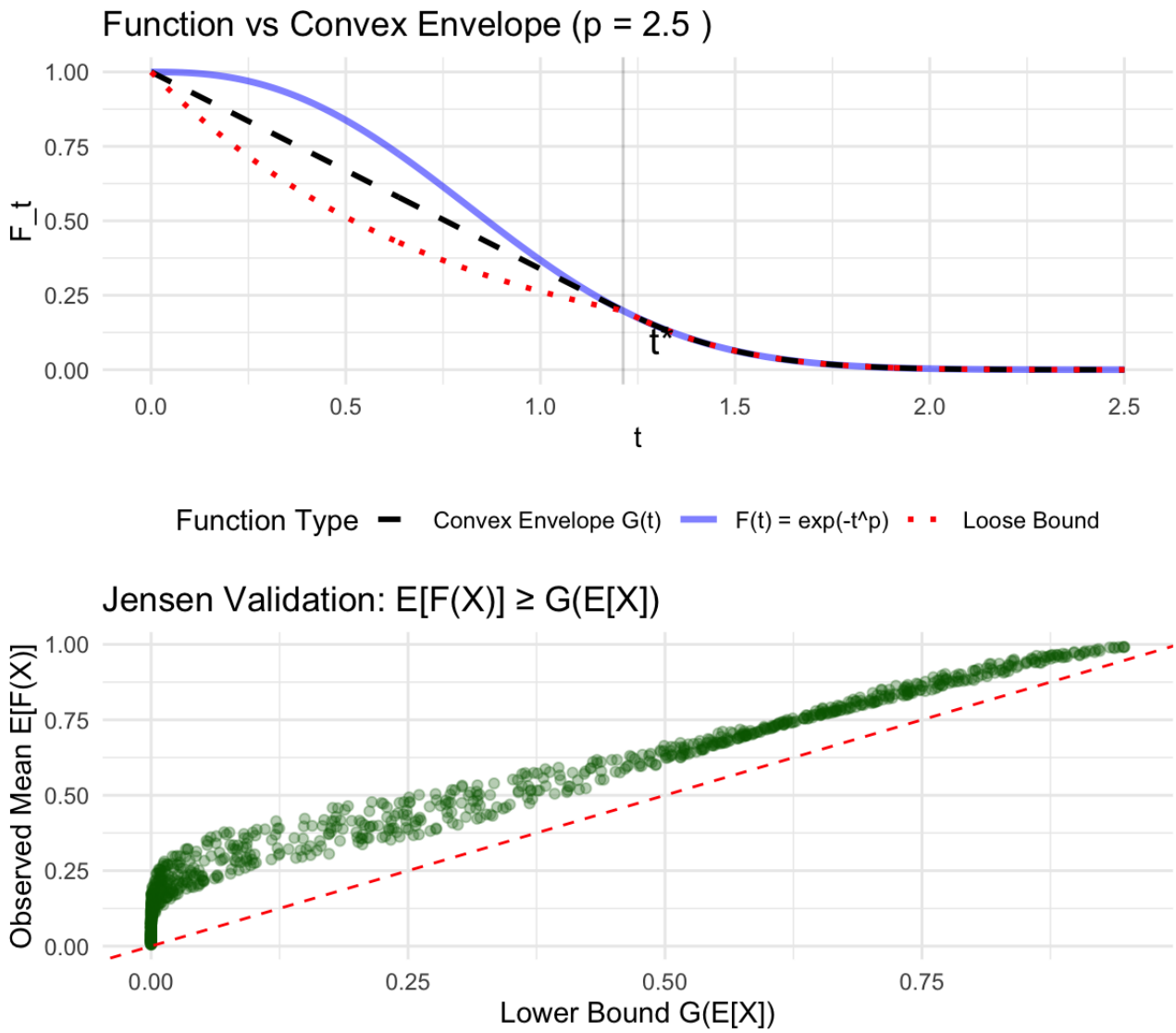
Suppose now that we want to handle 'the general case': I have an $\mathcal{X}$-valued random variables X, know the value of $\mathbf{E}[U(X)]$ for some $U : \mathcal{X} \to \mathbf{R}^K$, and I want to lower-bound $\mathbf{E}[V(X)]$ for some other $V : \mathcal{X} \to \mathbf{R}$. We follow a similar approach, seeking good convex lower bounds for V in terms of U, and this leads us to define (abusing notation slightly) $\left(V \circ U^{-1}\right)^\star : \mathbf{R}^K \to \mathbf{R}$ by

$$\left(V \circ U^{-1}\right)^\star(\alpha) := \sup\left\{\langle \alpha, U(x)\rangle - V(x) : x \in \mathcal{X}\right\}.$$

One can make this marginally more rigorous by interpreting $V \circ U^{-1}$ as an 'infimal projection', i.e. taking

$$\left(V \circ U^{-1}\right)(u) = \inf\left\{V(x) : U(x) = u\right\}.$$

Figure 1: The basic and simplified envelopes, and a Monte Carlo validation of the inequality.



Operationally, this perspective is useful because it implies the relationship

$$\langle \alpha, \mathbf{E}\left[U\left(X\right)\right]\rangle - \mathbf{E}\left[V\left(X\right)\right] \leq \left(V \circ U^{-1}\right)^{\star}\left(\alpha\right),$$

and so by optimising with respect to α, one obtains that

$$\mathbf{E}\left[V\left(X\right)\right] \geq \left(V \circ U^{-1}\right)^{\star\star}\left(\mathbf{E}\left[U\left(X\right)\right]\right),$$

where $\left(V \circ U^{-1}\right)^{\star\star}$ is the genuine convex dual of $\left(V \circ U^{-1}\right)^{\star}$. A bit of care shows that this relationship is effectively optimal, in the sense that without additional constraints on the nature of (X, U, V), the inequality cannot be sharpened further.